Findings Report

Operational Analytics & Financial Data Integrity Audit

> **Overview:** This report summarizes key insights derived from the database auditing framework across order subtotal reconciliation, delivery logistics performance, and customer lifetime value segmentation.

1. Financial Integrity & Subtotal Audit

Comparing order-level subtotals with calculated line-item sums (quantity × unit_price) allows us to isolate data discrepancies, floating-point rounding artifacts, and systematic billing errors. Non-active order states (Cancelled, Failed, Refunded) are excluded to ensure active revenue compliance.

Audit Category	Metric / Status	Threshold Filter	Business Impact
Subtotal Match	Passed Clean Audit	ABS(diff) <= 0.01	Verified transaction integrity
Precision Noise	Floating Point Artifacts	ABS(diff) > 0.01 Filtered	Eliminated false positive alerts
Active Orders	Excluded Cancelled/Refunded	WHERE status NOT IN (...)	Prevents distorted revenue metrics

2. Logistics Efficiency & Statistical Delivery Outliers

Delivery durations are tracked from order_date to delivery_date. Unusually slow deliveries are identified using a statistical threshold of Mean + (2 × Standard Deviation). This rule flags the upper tail (slowest ~2.5% of deliveries) as operational anomalies requiring vendor or courier review.

Fulfillment Metric	Calculation Basis	Statistical Threshold	Operational Action
Average Delivery Time	DATEDIFF / TIMESTAMPDIFF	Mean Duration	Establishes baseline SLA performance
Outlier Cutoff	Mean + 2 * Standard Deviation	Upper ~2.5% Tail	Isolates severe delays from daily variance
Platform Breakdown	GROUP BY Platform, City/State	Regional Segments	Identifies logistics & hub bottlenecks

3. Customer Value & Revenue Segmentation

A single-row customer report aggregates spending metrics, coupon engagement, and order frequency. Customers are categorized into segments based on total spending thresholds to drive targeted marketing campaigns.

Segment Name	Total Spend Threshold	Key Characteristics	Strategic Goal
High Value	>= 10,000	High lifetime spend, frequent purchases	VIP loyalty perks & concierge service
Medium Value	3,000 - 9,999	Consistent orders, moderate coupon use	Cross-sell incentives & repeat push
Low Value	< 3,000	Occasional buyers or single-order accounts	Re-engagement campaigns & discount offers